

Design and Implementation of a Database Management System for Restaurant Table Reservation and Food Service Management System

DBMS Project

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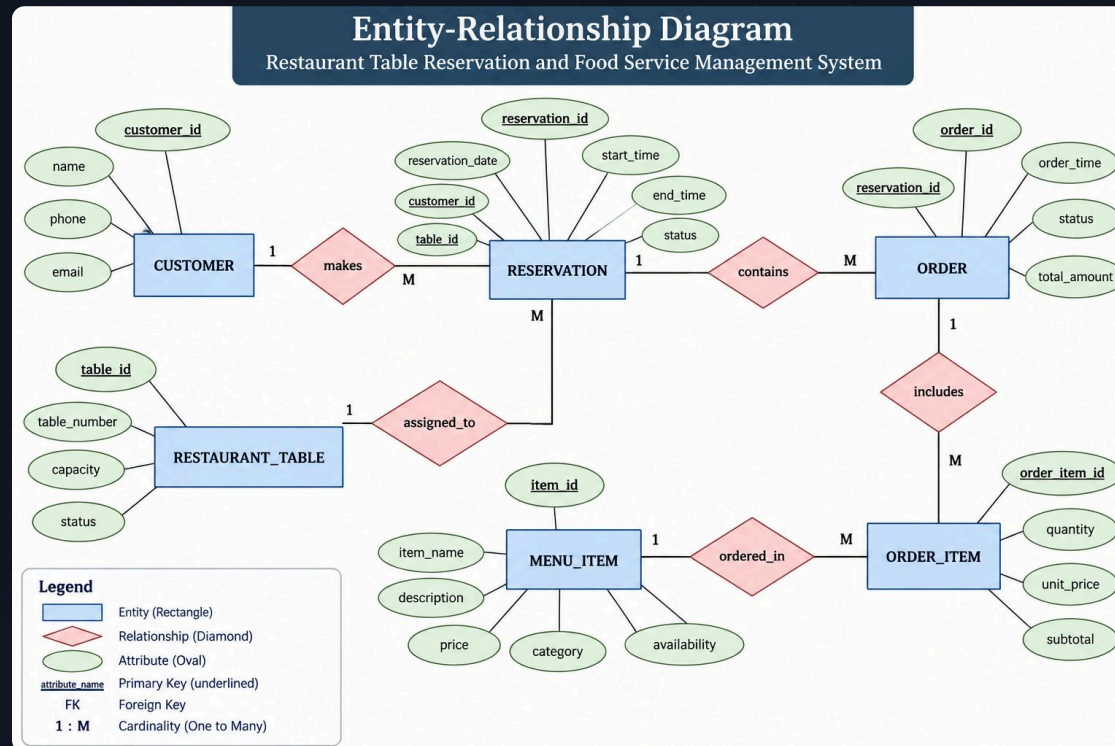
Roll Number: 25WU0102237

Subject: DBMS

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Conceptual Design – Entity Relationship Diagram

Conceptual representation of entities, attributes, relationships and cardinalities.



Relational Database Design

Field	Data Type	Key
customer_id	INT	PK
name	VARCHAR	
phone	VARCHAR	
email	VARCHAR	

Field	Data Type	Key
table_id	INT	PK
table_number	VARCHAR	
capacity	INT	
status	VARCHAR	

Relational Database Design

Field	Data Type	Key
reservation_id	INT	PK
customer_id	INT	FK
table_id	INT	FK
reservation_date	DATE	
start_time	TIME	
end_time	TIME	
status	VARCHAR	

Relational Database Design

Field	Data Type	Key
order_id	INT	PK
reservation_id	INT	FK
order_time	TIME	
status	VARCHAR	
total_amount	DECIMAL	

Relational Database Design

Field	Data Type	Key
item_id	INT	PK
item_name	VARCHAR	
description	VARCHAR	
price	DECIMAL	
category	VARCHAR	
availability	VARCHAR	

Relational Database Design

Field	Data Type	Key
order_item_id	INT	PK
order_id	INT	FK
item_id	INT	FK
quantity	INT	
unit_price	DECIMAL	
subtotal	DECIMAL	

Filled-In RDBMS Tables

item_id	item_name	category	price	availability
m01	Margherita Pizza	Main Course	300.00	AVAILABLE
m02	Creamy Alfredo Pasta	Main Course	250.00	AVAILABLE
m03	Sizzling Chocolate Brownie	Dessert	150.00	AVAILABLE

order_id	reservation_id	order_time	status	total_amount
ord901	r101	19:15	PREPARING	450.00

order_item_id	order_id	item_id	quantity	unit_price	subtotal
oi01	ord901	m01	1	300.00	300.00
oi02	ord901	m03	1	150.00	150.00

Filled-In RDBMS Tables

CUSTOMER

customer_id	name	phone	email
c01	Rahul Verma	9876543210	rahul@gmail.com
c02	Ananya Roy	9123456780	ananya@gmail.com
c03	Karthik Nair	9988776655	karthik@gmail.com

RESTAURANT_TABLE

table_id	table_number	capacity	status
t01	T-01	2	AVAILABLE
t02	T-02	4	RESERVED
t03	R-01	6	AVAILABLE

Filled-In RDBMS Tables

RESERVATION

reservation_ id	customer_ id	table_ id	reservation_date	start_time	end_time	status
r101	c01	t02	2026-09-16	19:00	20:30	CONFIRMED
r102	c02	t01	2026-09-16	20:00	21:00	CONFIRMED

Query Execution

This section demonstrates the execution of query within our database management system.